

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM		
SYSTEM:	Emergency Diesel Generator		
TASK:	Start and synchronize an Emergency Diesel Generator		
TASK NUMBER:	113 001 05 01		
JPM NUMBER:	11-01 NRC IP-j		
ALTERNATE PATH:	☐	K/A NUMBER:	064 A2.09
APPLICABILITY:	IMPORTANCE FACTOR:		
EO ☐	RO ☒	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:		In Plant / Simulate	
REFERENCES:	2A Diesel Generator Operation, S2.OP-SO.DG-0001(Q)Rev. 37 (checked 08-13-12)		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	<u>20 min</u>		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	<u>N/A</u>		
Developed By:	G Gauding Instructor	Date:	08-13-12
Validated By:	Rydell SME or Instructor	Date:	8-23-12
Approved By:	Training Department	Date:	
Approved By:	Operations Department	Date:	
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

SYSTEM: EDG

TASK: Start and synchronize an Emergency Diesel Generator

**TASK
NUMBER:** 113 001 05 01

**INITIAL
CONDITIONS:** Unit 2 is operating at 100% power, with no equipment OOS. Engineering has requested a loaded run of 2A EDG.

INITIATING CUE:

The Unit 2 CRS has directed you to locally start and load 2A D/G IAW S2.OP-SO.DG-0001(Q), Section 5.2. & 5.4

Prerequisites, Precautions and Limitations, and Section 5.1, Diesel Generator Startup Checks have been performed.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).

Task Standard for Successful Completion:

Perform S2.OP-SO.DG-0001 Sections 5.2 and 5.4 in correct order and to completion.

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TASK: Locally start and load an Emergency Diesel Generator

NAME: _____
DATE: _____

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
		Provide marked up copy of S2.OP-SO.DG-0001 2A Diesel Generator Operation			
	5.2.1	NOTIFY NCO that 2A Diesel Generator is to be locally started.	Locates nearest page. Cue: NCO acknowledges.		
	5.2.2	CHECK voltage permissive indicator light 2DAE4-LT2, EDG VOLTAGE, on Generator Control Panel is OFF	Points out voltage permissive indicator light 2DAE4-LT2, EDG VOLTAGE, on Generator Control Panel. Cue: Light is off		
	5.2.3	CHECK speed permissive indicator light 2DAE4-LT3, EDG SPEED, on Generator Control Panel is OFF.	Points out speed permissive indicator light 2DAE4-LT3, EDG SPEED, on Generator Control Panel. Cue: Light is off		
*	5.2.4	PLACE 2A-DF-GCP-1, 2A DIESEL GEN LOADING SW in MANUAL (DROOP).	Points out 2A-DF-GCP-1, 2A DIESEL GEN LOADING SW., and simulates rotating to DROOP position. Cue: Annunciator B-9, Generator loading in Droop Mode annunciates.		
	5.2.5	ENSURE B-9, GENERATOR LOADING IN DROOP MODE, is in alarm.	Acknowledges alarm B-9, Generator Loading in Droop Mode, and simulates depressing Silence PB. Cue: Alarm B-9, Generator Loading in Droop Mode has silenced.		

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	5.2.6	PLACE 2A-DF-SS, 2A DG STOP/START SWITCH in START.	Points out 2A-DF-SS, 2A DG STOP/START SWITCH, and simulates placing in START. Cue: 2A Diesel is accelerating.		
*	5.2.7	<u>IF</u> Diesel Generator Speed is <u>NOT</u> 900 rpm, <u>THEN</u> SET speed to 900 rpm using the SPEED CONTROL SWITCH (GS).	Cue when Speed indicator is checked: Speed is 880 rpm. Points out SPEED CONTROL SWITCH (GS) and simulates turning in RAISE direction. Cue: 2A Diesel Speed is 900 rpm.		
	5.2.8	<u>IF</u> 2AD1AX6D*, 2A Diesel Generator 125VDC breaker is closed, THEN: - CHECK voltage permissive indicator light 2DAE4-LT2, EDG VOLTAGE, on Generator Control Panel is ON. - CHECK speed permissive indicator light 2DAE4-LT3, EDG SPEED, on Generator Control Panel is ON.	Points out voltage permissive indicator light 2DAE4-LT2, EDG VOLTAGE. Cue: Voltage permissive indicator light 2DAE4-LT2, EDG VOLTAGE Light is on Points out speed permissive indicator light 2DAE4-LT3, EDG SPEED Cue: Speed permissive indicator light 2DAE4-LT3, EDG SPEED light is on		

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	5.2.9	IF Field Ground Relay 64/G white indicating light is OFF, <u>AND</u> C-6, GENERATOR FIELD GROUND, is clear, <u>THEN</u> : A. RESET 64/G relay B. ENSURE 64/G white indicating light is illuminated.	Points out Field Ground Relay 64/G white indicating light. Cue: Field Ground Relay 64/G white indicating light is illuminated.		
	5.2.10	ENSURE 2A Diesel Generator K1C Field Flashing Relay Supervisory Light is OFF.	Points out 2A Diesel Generator K1C Field Flashing Relay Supervisory Light Cue: 2A Diesel Generator K1C Field Flashing Relay Supervisory Light is off.		
	5.2.11	RECORD the following Diesel Generator Start Readings:	Points out gages and switches to determine readings. Cue when proper gauge is pointed out : <u>2VM189</u> Gen. Volts - 4160 Volts on all 3 phases <u>2FM186</u> Gen. Frequency - 60 Hz <u>2PL6429</u> LO Hdr. Press - 80 psig <u>2PL6449</u> JW Hdr. Press - 45 psig <u>2PL7209</u> Air Manifold Press - 0 psig <u>2TA16524</u> Gen. Stator Temp - 187°		

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	5.2.12	IF 2A Diesel Generator is to be operated unloaded for an extended period of time(>30 minutes), <u>THEN</u> INITIATE Section 5.7, Diesel Generator Running Checks.	Cue: Diesel will be parallel loaded.		
	5.2.13	IF Diesel Generator is to be parallel loaded, <u>THEN</u> INITIATE Section 5.4, Diesel Generator Parallel Loading.			
	5.4.1	IF Section 5.1 Diesel Generator Startup Checks, was <u>NOT</u> performed, <u>THEN</u> at 2A DG 4KV cabinet cubicle breaker 2AD1AX6D, PLACE selector switch 2A DG SYNC ENABLE in ENABLE.	Verifies Section 5.1 is complete.		
	5.4.2	NOTIFY NCO that 2A EDG is to be synchronized and loaded locally.	Cue: NCO acknowledges 2A DG is to be synchronized and loaded locally.		
	5.4.3	ENSURE 2A-DF-GCP-1, 2A DIESEL GEN LOADING SW, in MANUAL (DROOP)	Verifies 2A-DF-GCP-1, 2A DIESEL GEN LOADING SW, is in MANUAL (DROOP)		

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*	5.4.4	ADJUST 2A EDG output voltage, as indicated on VOLTMETER-GEN (2VM189) to 50 - 100 Volts higher than that of 2A 4KV Vital Bus voltage, as indicated by VOLTMETER-BUS (2VM190), using the VOLTAGE CONTROL SWITCH (VCS).	When operator points out 2VM189 and 2VM190, then Cue: 2VM189 reads 4150, and 2VM190 reads 4150. Points out VOLTAGE CONTROL SWITCH (VCS) and simulates going to RAISE. Cue: 2VM189 reads 4210, and 2VM190 reads 4150.		
	5.4.5	ENSURE generator terminal voltage is present on all 3 phases by rotating VOLTMETER SWITCH - GEN (VS-G) through each position <u>AND</u> OBSERVING voltmeter 2VM189.	Points out VOLTMETER SWITCH - GEN (VS-G), simulates rotating through each position, while observing 2VM189 Cue: Voltage is present on all three phases.		
	5.4.6	ENSURE 2A 4KV Vital Bus voltage is present on all 3 phases by rotating VOLTMETER SWITCH - BUS (VS-B) through each position <u>AND</u> OBSERVING voltmeter 2VM190.	Points out VOLTMETER SWITCH - BUS (VS-B), simulates rotating through each position, while observing 2VM190 Cue: Voltage is present on all three phases.		

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	5.4.7	PLACE the following switches in "1-2": - VOLTMETER SWITCH - GEN (VS-G) - VOLTMETER SWITCH - BUS (VS-B)	Points out VOLTMETER SWITCH - GEN (VS-G) and VOLTMETER SWITCH - BUS (VS-B) Cue: Switches are in 1-2.		
	5.4.8	ENSURE 2DAE4-LT2, EDG Voltage indication light is ON.	Points out 2DAE4-LT2, EDG Voltage indication light. Cue: 2DAE4-LT2, EDG Voltage indication light is on.		
	5.4.9	ENSURE 2DAE4-LT3, EDG Speed indication light is ON.	Points out 2DAE4-LT3, EDG Speed indication light. Cue: 2DAE4-LT2, EDG Voltage indication light is on.		
	5.4.10	SYNCHRONIZE 2A Diesel Generator to 2A 4KV Vital Bus as follows:			
*	A.	PLACE 2A-DF-SYNCH, 2A DG SYNC SWITCH (SS) to ON.	Points out 2A-DF-SYNCH, 2A DG SYNC SWITCH (SS), and simulates placing in ON position.		

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	F	COORDINATE the following when the synchroscope is at "12 o'clock" (+zero, -2 minutes): 1. CLOSE 2A-DF-GCP-3 GENERATOR CIRCUIT BREAKER SWITCH (BCS) 2. RAISE 2A Diesel Generator load to $\geq 500\text{KW}$ using SPEED CONTROL SWITCH (GS) to prevent tripping the Diesel Generator Breaker on reverse power.	Points out 2A-DF-GCP-3 GENERATOR CIRCUIT BREAKER SWITCH (BCS) and simulates taking to the CLOSE position when the synch scope is at 12 o'clock. Cue 2A Diesel Generator Breaker has closed. Points out SPEED CONTROL SWITCH (GS) simulates going to RAISE while observing KW meter. Cue Load is 500 KW.		
	G	PLACE 2A-DF-SYNCH, 2A DG SYNCH SWITCH (SS) to OFF.	Points out 2A-DF-SYNCH, 2A DG SYNCH SWITCH (SS), and simulates placing it in OFF position. Cue: JPM is complete.		

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*	B.	ADJUST Diesel speed using the SPEED CONTROL SWITCH (GS) such that Synchroscope pointer rotates slowly in the FAST (clockwise) direction.	When operator points out synchroscope, then Cue: Scope is going in the slow direction. Points out SPEED CONTROL SWITCH (GS) and simulates going to RAISE. Cue Synchroscope pointer rotating slowly in the FAST direction.		
	C.	ENSURE the following four SYNC CHECK RELAY 25 UPPER AND LOWER VOLTAGE LIMIT LEDS are ON. "Upper Voltage Limit" "L OK" "B OK" "Lower Voltage Limit" "L OK" "B OK"	Points out four SYNC CHECK RELAY 25 UPPER AND LOWER VOLTAGE LIMIT LEDS. Cue as each LED is located: Light is on.		
	D.	<u>IF</u> the SYNC CHK RELAY 25 ΔF OK LED is OFF, <u>THEN</u> ADJUST 2A Diesel Generator speed using the SPEED CONTROL SWITCH (GS) until the ΔF OK LED is ON.	Points out SYNC CHK RELAY 25 ΔF OK LED. Cue Light is on.		
	E.	ENSURE DG SYNC PERMISSIVE green indicating light is ON each time the synchroscope indicator is near 12 o'clock position (+/- approximately 3 minutes), <u>AND</u> is OFF in any other position of the synchroscope.	Cue DG SYNC PERMISSIVE green indicating light is ON each time the synchroscope indicator is near 12 o'clock position (+/- approximately 3 minutes), <u>AND</u> is OFF in any other position of the synchroscope.		

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JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- _____ 1. Task description and number, JPM description and number are identified.
- _____ 2. Knowledge and Abilities (K/A) references are included.
- _____ 3. Performance location specified. (in-plant, control room, or simulator)
- _____ 4. Initial setup conditions are identified.
- _____ 5. Initiating and terminating Cues are properly identified.
- _____ 6. Task standards identified and verified by SME review.
- _____ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- _____ 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. _____ Date _____
- _____ 9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: _____

Date: _____

SME/Instructor: _____

Date: _____

SME/Instructor: _____

Date: _____

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

INITIAL CONDITIONS:

Unit 2 is operating at 100% power, with no equipment OOS. Engineering has requested a loaded run of 2A EDG.

INITIATING CUE:

The Unit 2 CRS has directed you to locally start and load 2A D/G IAW S2.OP-SO.DG-0001(Q), Section 5.2. & 5.4

Prerequisites, Precautions and Limitations, and Section 5.1, Diesel Generator Startup Checks have been performed.